
The Risk of Developing Breast Cancer Within the Next 5, 10, or 20 Years of a Woman's Life

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Background: The lifetime risk of developing breast cancer is a frequently misinterpreted statistic. Risk projections over a shorter time period, conditioned on current age, may be less prone to misconceptions and more relevant to populations at different ages. The purpose of this study was to estimate the risk of developing breast cancer within the next 5, 10, or 20 years for women currently aged 30 to 70 years in California's four major race/ethnic groups.

Methods: Life tables were used to obtain risk estimates based on 1993–1997 breast cancer incidence rates from the California Cancer Registry and statewide mortality rates.

Results: For women currently aged 50, the estimated risk of developing invasive breast cancer within 5 years varied from 0.8% (1 in 133) among Hispanics to 1.3% (1 in 75) among Caucasians. Risk estimates within 10 years were 2.9% (1 in 34) among Caucasians, 2.3% (1 in 43) among African Americans, 2.0% (1 in 51) among Asian/Pacific Islanders, and 1.6% (1 in 63) among Hispanics. Within 20 years, estimated risks increased to 6.6% (1 in 15) among Caucasians, 5.0% (1 in 20) among African Americans, 3.9% (1 in 26) among Asian/Pacific Islanders, and 3.7% (1 in 27) among Hispanics. Risk estimates were 8% to 20% higher when in situ tumors were included in the calculations.

Conclusions: Based on these estimates, the baseline risk of developing breast cancer in the next 1 or 2 decades of life varies by race/ethnicity and current age, but may be lower than the risk perceived by most women.

Medical Subject Headings (MeSH): breast neoplasms, ethnic groups, life tables, racial stocks, risk (Am J Prev Med 2001;20(3):214–218)

Introduction

The lifetime risk of developing breast cancer, frequently translated into the familiar statistic of one-in-nine women, is commonly cited in the health care community and the media. Notwithstanding the value of lifetime risk estimates as a way to communicate the need for breast cancer screening, misconceptions about this frequently cited number may be generating excessive fear and anxiety among women. For example, in a recent population-based survey, most respondents overestimated their average lifetime risk of breast cancer, independent of their age or prior exposure to information on breast cancer.¹ A community survey of mammography use found that 21% of women in their 40s perceived their personal risk as "higher than most."² Another study on perceptions of breast cancer risk found that 37% of women aged

<50 overestimated their risk of developing breast cancer within the next 10 years by a factor of ten or more.³ The authors suggested that the one-in-nine lifetime risk may have been misinterpreted by women as being a short-term probability.³ Counseling apparently did not prevent women from overestimating their risks. In a randomized trial of risk counseling among first-degree relatives of breast cancer patients, about two thirds of women who had been counseled continued to overestimate their lifetime risks substantially.⁴ Similarly, many women with a family history of breast cancer who attended genetic counseling failed to recall risk figures correctly, and 66% could not accurately recall their own lifetime chance of developing the disease.⁵

Lifetime risk is defined as a newborn female's probability of eventually being diagnosed with breast cancer. It assumes that age-specific breast cancer incidence rates and mortality rates from all causes other than breast cancer will remain stable over her lifetime period. Risk projections over a shorter time period are therefore more likely to be accurate, and may be more meaningful to the general public. In addition, because breast cancer incidence and mortality vary markedly by race/ethnicity,^{6,7} race-specific risk estimates are more

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Table 1. Average annual population estimates^a and female breast cancer 5-year age-specific incidence rates,^b by age group and race/ethnicity,^c California, 1993–1997

Age group	Population estimates				Incidence rates (per 100,000 women)							
					Invasive tumors only				Invasive and in situ tumors			
	African American	Asian/PI	Caucasian	Hispanic	African American	Asian/PI	Caucasian	Hispanic	African American	Asian/PI	Caucasian	Hispanic
0–4	105,555	143,545	524,843	616,276	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5–9	100,566	129,644	536,355	469,043	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10–14	87,115	123,966	496,004	387,904	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
15–19	81,599	116,386	450,913	344,454	0.0	0.0	0.1	0.2	0.0	0.0	0.1	0.2
20–24	81,970	125,240	477,303	354,539	2.4	1.3	1.3	1.6	2.7	1.4	1.4	1.6
25–29	92,684	132,472	579,448	398,805	11.2	5.3	6.9	6.4	11.0	5.7	7.4	6.8
30–34	103,752	149,935	722,708	397,281	35.7	20.0	22.8	19.5	37.8	22.4	24.7	20.1
35–39	100,822	157,788	767,232	348,393	60.9	51.1	54.9	42.0	67.3	57.7	62.5	46.0
40–41	86,779	144,929	714,183	273,943	118.9	103.2	115.4	82.4	132.8	124.1	135.9	92.9
45–49	70,103	118,715	640,422	203,253	181.7	151.3	200.7	126.8	206.6	180.6	238.3	142.8
50–54	52,762	82,352	511,483	146,904	204.7	186.3	267.8	150.4	243.7	222.7	318.0	173.5
55–59	43,198	66,086	409,220	114,386	274.6	207.0	318.2	175.4	314.4	246.0	373.9	196.9
60–64	35,926	58,008	370,966	95,868	304.0	201.7	382.7	210.9	347.9	234.5	440.2	236.4
65–69	31,120	53,501	384,459	83,491	331.0	210.5	438.0	249.9	390.1	248.2	505.9	285.1
70–74	27,296	43,664	384,039	63,346	339.2	202.5	488.7	266.2	394.2	231.8	556.3	301.5
75–79	19,731	28,986	304,441	41,801	380.1	213.2	504.3	272.2	432.8	251.8	563.9	301.4
80–84	13,335	16,981	221,780	28,762	357.0	193.2	487.6	221.8	403.4	221.4	530.4	246.2
85+	10,750	12,608	199,168	26,418	351.6	145.9	407.8	165.8	374.0	149.1	430.3	177.9

^aAverage annual population estimates for the 1993–1997 period, California Department of Finance.

^bIncidence rates are based on the number of first primary breast cancers diagnosed in California from 1993 through 1997, and reported to the California Cancer Registry as of November 1999.

^cRace/ethnicity categories are mutually exclusive. Women of Hispanic ethnicity were identified by medical record and/or surname, and may be of any race.

Asian/PI, Asian/Pacific Islander

informative than estimates based on rates for women of all race/ethnic groups combined. The purpose of this study was to estimate the baseline risk of being diagnosed with breast cancer within the next 5, 10, or 20 years of life for California women of four different race/ethnicities: non-Hispanic Caucasian, Hispanic, non-Hispanic African American, and non-Hispanic Asian/Pacific Islander (PI). Risk estimates were based on cancer incidence data from the California Cancer Registry (CCR), California's statewide population-based cancer surveillance system, and mortality data from the Center of Health Statistics, California Department of Health Services. The unique diversity of the California population allowed for stable estimates of risk for these four race/ethnic groups.

Methods

In this report, the risk of developing cancer was taken to mean the risk of being diagnosed with cancer. Estimates of risk were obtained following the methodology of Feuer et al.,^{8,9} based on breast cancer incidence and mortality rates from all causes other than breast cancer in California from 1993 to 1997. Two sets of estimates were calculated: (1) for all breast cancers (invasive plus in situ tumors) and (2) for invasive breast cancer only. The SAS System release 6.12 (SAS Institute Inc., Cary, NC) and DEVCAN, a computer program available through the National Cancer Institute (NCI),⁸ were used for all data analyses.

Breast Cancer Incidence and Mortality Rates

During the 1993–1997 period, 109,165 female breast cancers were diagnosed in California and reported to the CCR as of November 1999. Age-specific incidence rates for breast cancer (Table 1) and mortality rates for all causes other than breast cancer were calculated for each race/ethnic group. Incidence rates were based on the number of first primary breast cancers (invasive and invasive plus in situ cases) diagnosed in California women from 1993 to 1997. Non-breast-cancer mortality rates were obtained by subtracting breast cancer related deaths from the total number of deaths occurring in California during the period (Center of Health Statistics, California Department of Health Services). Race- and age-specific population estimates were obtained from the Demographic Research Unit, California Department of Finance.

Definition of Race/Ethnicity

Race/ethnicity for breast cancer cases, deaths, and population estimates was grouped into four mutually exclusive categories of non-Hispanic Caucasian, non-Hispanic African American, Hispanic, and non-Hispanic Asian/Pacific Islander. Hispanic ethnicity of cases and deaths was based on information on the medical record or death certificate. In addition, women of any race who had a last name (or maiden name, when present) appearing on the 1980 U.S. Census list of 12,497 Hispanic surnames were also classified as Hispanic. The use of information on medical records together with surnames to identify Hispanics has been shown to improve

Table 2. Risk of being diagnosed with breast cancer within the next 5, 10, or 20 years, by race/ethnicity,^a assuming no breast cancer at current age: California, 1993–1997

Current age	Invasive tumors only						Invasive or in situ tumors					
	+5 years		+ 10 years		+20 years		+5 years		+ 10 years		+20 years	
	Risk (%)	One in:	Risk (%)	One in:	Risk (%)	One in:	Risk (%)	One in:	Risk (%)	One in:	Risk (%)	One in:
All races												
30	0.1	895	0.4	250	1.8	56	0.1	836	0.4	250	2.0	50
40	0.5	189	1.4	71	3.8	26	0.6	162	1.6	63	4.5	22
50	1.2	86	2.5	40	5.7	18	1.4	73	3.0	33	6.7	15
60	1.6	61	3.4	29	7.0	14	1.9	53	4.0	25	8.0	13
70 ^b	2.1	47	4.1	24	—	—	2.5	40	4.7	21	—	—
African American												
30	0.2	563	0.5	200	1.9	53	0.2	532	0.5	200	2.1	48
40	0.6	169	1.5	67	3.7	27	0.7	152	1.7	59	4.2	24
50	1.0	98	2.3	43	5.0	20	1.2	83	2.7	37	5.8	17
60	1.5	67	3.0	33	5.5	18	1.7	58	3.5	29	6.4	16
70 ^b	1.6	61	3.2	31	—	—	1.9	52	3.7	27	—	—
Asian/Pacific Islander												
30	0.1	1001	0.4	250	1.6	63	0.1	894	0.4	250	1.9	53
40	0.5	194	1.3	77	3.2	31	0.6	161	1.5	67	3.8	26
50	0.9	107	2.0	51	3.9	26	1.1	89	2.3	43	4.6	22
60	1.0	98	2.0	50	3.9	26	1.2	84	2.4	42	4.5	22
70 ^b	1.0	98	2.0	50	—	—	1.2	85	2.3	43	—	—
Caucasian												
30	0.1	879	0.4	250	1.9	53	0.1	811	0.4	250	2.3	43
40	0.6	174	1.6	62	4.4	23	0.7	148	1.9	53	5.2	19
50	1.3	75	2.9	34	6.6	15	1.6	63	3.4	29	7.7	13
60	1.9	52	4.0	25	8.0	13	2.2	45	4.7	21	9.2	11
70 ^b	2.5	41	4.7	21	—	—	2.8	35	5.3	19	—	—
Hispanic												
30	0.1	1026	0.3	333	1.3	77	0.1	995	0.3	333	1.5	67
40	0.4	243	1.0	100	2.6	38	0.5	216	1.2	83	3.0	33
50	0.8	133	1.6	63	3.7	27	0.9	115	1.8	56	4.2	24
60	1.1	95	2.2	45	4.5	22	1.2	84	2.5	40	5.0	20
70 ^b	1.3	76	2.5	40	—	—	1.5	67	2.8	36	—	—

^aRace/ethnicity categories are mutually exclusive. Women of Hispanic ethnicity were identified by medical record and/or surname, and may be of any race.

^bEstimates were truncated at age 85, and are, therefore, not available for age 70 plus 20.

the classification of Hispanic ethnicity, which is often under-reported in medical records and death certificates.¹⁰

Risk Estimates

Calculations were based on a hypothetical cohort of 10 million newborn females, for whom standard multiple-decrement life tables were derived.^{8,9} During each 5-year age interval, females in this cohort were considered to be at risk for two mutually exclusive events: (1) developing breast cancer for the first time in their lives or (2) dying of other causes without having developed breast cancer. Mortality among women with breast cancer did not affect the calculations, since they already had developed breast cancer. Probabilities derived from statewide incidence and mortality rates were applied to the hypothetical cohort to generate the number of women expected to develop breast cancer or to die of other causes without having developed breast cancer, during each age interval. Women who survived the interval

free of breast cancer became the population at risk at the beginning of the next age interval. The risk of developing breast cancer within a particular age interval was estimated as the number of expected incident cancers developing during that interval divided by the population at risk (free of breast cancer) at the beginning of that period.

Results

Estimates of a woman's risk of being diagnosed with invasive breast cancer within the next 5, 10, or 20 years (assuming she is free of breast cancer at her current age) are shown in Table 2. These estimates reflect the variation in the incidence of breast cancer by race/ethnicity in California, where rates have been highest among Caucasian women and lowest among Hispanic or Asian/PI women.⁶ For example, based on 1993–1997

data, the estimated risk of developing invasive breast cancer for a 50-year-old woman of Hispanic ethnicity was 0.8% (1 in 133) within the next 5 years, 1.6% (1 in 63) within the next 10 years, and 3.7% (1 in 27) within the next 20 years. For a Caucasian woman of the same age, the risk of invasive breast cancer during the same period was substantially higher: 1.3% (1 in 75) within 5 years, 2.9% (1 in 34) within 10 years, and 6.6% (1 in 15) within 20 years. A 50-year-old African-American woman had a 1.0% (1 in 98) risk of developing invasive breast cancer by age 55, a 2.3% (1 in 43) risk by age 60, and a 5.0% (1 in 20) risk by age 70. Similarly, the estimated risks for 50-year-old Asian/PI women were 0.9% (1 in 107), 2.0% (1 in 51), and 3.9% (1 in 26) within the next 5, 10, and 20 years, respectively.

The older the woman, the higher the risk of developing breast cancer in the next 5 to 20 years of her life, reflecting the fact that incidence rates increase with age.⁶ However, the risk in the next 20 years for a woman at any age was always higher than the risk in the next 10 years for a woman 10 years older. For example, the estimated risk by age 70 for a 50-year-old woman was higher than the risk by age 70 for a 60-year-old woman. This is because, unlike an older woman who has already survived and reached her current age free of breast cancer, a woman 10 years younger is still at risk of developing breast cancer in the next decade of her life.

Overall, risk estimates increased by 8% to 20% when in situ cancers were included in the calculations (Table 2). As expected, such an increase in risk estimates for the age groups included in this study paralleled the proportion of in situ tumors among all breast cancers diagnosed in California from 1993 to 1997.⁶

Discussion

The lifetime risk of developing invasive breast cancer, albeit a powerful translational statistic, has been frequently misinterpreted by the lay media and the public. The commonly cited one-in-eight figure (or one in nine when truncated to age 85) represents a newborn female's likelihood of eventually being diagnosed with invasive breast cancer, and does not apply to women of all ages. For women who are currently over 40, estimates of the risk of developing invasive breast cancer at some time during the remainder of their lives actually decrease with age,^{6,11} and differ substantially from estimates of lifetime risk for a newborn. The difference in risk is due to the fact that an older woman has already survived several decades free of breast cancer, while a newborn is still at risk of developing breast cancer (or dying of other causes) until she reaches that particular age. Another frequent misconception is that lifetime risk estimates are valid only if all women live to a specific age (e.g., 85 or 95 years), when in reality calculations take into account the likelihood of dying of other causes at any age.⁹ In addition to these miscon-

ceptions, risk estimates are projections of current rates into the future and are, therefore, valid only if these rates remain stable over time. In the short term, such an assumption may hold true, but is probably not realistic over a long period of time. On one hand, improvements in screening technology may inflate incidence rates, since more tumors are detected at an earlier stage of development. On the other hand, breast cancer research may lead to more-effective prevention strategies and, therefore, to an actual decrease in risk. For these reasons, risk estimates based on current age and calculated for the next decade or two may be more accurate, less prone to misconceptions and, therefore, more meaningful to the general population.

In this study, the risk of developing breast cancer varied markedly by race/ethnicity, reflecting the already recognized race/ethnic differences in breast cancer rates in California and the United States.^{6,7,11} The accuracy of ethnic classification is an ever-present concern in health studies that utilize medical record data, since ethnicity assignment in medical records may involve subjective appraisals by hospital personnel. The possibility of some misclassification of Hispanic ethnicity in this study cannot be ruled out. However, such misclassification was minimized by supplementing medical record information with use of the patient's surname. An earlier California study demonstrated that use of medical records plus surname reduced the bias observed with use of either method alone when compared with self-identification of ethnicity.¹⁰

Differences in mortality rates and in screening practices may have a substantial impact on risk estimates. High non-breast-cancer-related mortality rates decrease the number of women in the life table at risk for breast cancer. When cancer probabilities are applied to a smaller population at risk, the expected number of breast cancers will be accordingly smaller. Assuming identical incidence rates, risk estimates would be lower for a race/ethnic group with higher mortality rates, because the number of cancers in the life table (to be divided by the initial population at risk) would be smaller. Mammography screening can also affect breast cancer risk in two opposite ways. Screening may reduce risk because prevalent cases are excluded from the population at risk, but it may also increase the risk due to lead time and, particularly for in situ tumors, over-diagnosis bias.

This study is based on California statewide data, and we do not know to what extent these risk estimates can be extrapolated to other geographic areas in the United States. However, interval risks of developing invasive breast cancer for African-American and Caucasian women in this study are consistent with estimates from the NCI's Surveillance, Epidemiology, and End Results (SEER) program, based on data from 1995 through 1997.¹¹ Risk estimates for Hispanic and Asian/PI women, or for invasive/in situ tumors were

not available from SEER or, to the best of our knowledge, from other population-based registries.

The population-based estimates of risk presented in this report do not take into account risk factors other than age and race/ethnicity, and represent the average risk of breast cancer. More-complex models have been developed that project the risk of breast cancer with consideration of individual risk factors.^{12–14} One of these models, developed by Claus et al.,¹² calculates breast cancer risk for women with a first-degree family history of ovarian cancer. The Gail model,¹³ and a modification of this model developed by the National Surgical Adjuvant Breast and Bowel Project,¹⁴ account for individual risk factors such as current age, age at menarche, age at first live birth, number of previous breast biopsies, presence of atypical hyperplasia, and number of affected first-degree relatives. Such models can be very useful to counsel individual women, but have limited use for communicating risk to a broader population. In addition, these models were developed based on the occurrence of breast cancer in Caucasian women only, and their validity for women of other race/ethnic groups has not been determined.¹⁵

Clearly, age and race/ethnicity alone cannot predict the risk of developing cancer for an individual woman, for whom the limitations of risk projections should be recognized. However, population-based estimates can be used by health care providers in primary care settings to increase awareness and encourage screening among older women and, at the same time, put breast cancer risk into perspective among younger women. Based on these estimates, the baseline risk of developing breast cancer in the next 1 or 2 decades of life may be considerably lower than the risk perceived by most women. Use of this statistic to communicate a woman's average risk of developing breast cancer at her current

age may reduce the current confusion and anxiety around this serious issue.

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